(e) Magnetic susceptibility

When a substance is kept in a magnetising field $\vec{H}$, magnetic susceptibility gives information about how a material respond to the external (applied) magnetic field. In other words, the magnetic susceptibility measures, how easily and how strongly a material can be magnetised. It is defined as the ratio of the intensity of magnetisation $(\vec{M})$ induced in the material due to the magnetising field $(\vec{H})$

$$\chi_m = \frac{|\vec{M}|}{|\vec{H}|} \qquad (3.36)$$

It is a dimensionless quantity. For an isotropic medium, susceptibility is a scalar but for non-isotropic medium, susceptibility is a tensor. Magnetic susceptibility for some of the isotropic substances is given in Table 3.1.

Table 3.1 Magnetic susceptibility for various materials

Material	Magnetic susceptibility (χ_m)
Aluminium	2.3×10^{-5}
Copper	-0.98×10^{-5}
Diamond	-2.2×10^{-5}
Gold	-3.6×10^{-5}
Mercury	-3.2×10^{-5}
Silver	-2.6×10^{-5}
Titanium	7.06×10^{-5}
Tungsten	6.8×10^{-5}
Carbon dioxide (1 atm)	-2.3×10^{-9}
Oxygen (1 atm)	2090×10^{-9}

EXAMPLE 3.10

Compute the intensity of magnetisation of the bar magnet whose mass, magnetic moment and density are 200 g, 2 A m^2 and 8 g cm^{-3}, respectively.

Solution

Density of the magnet is

$$Density = \frac{Mass}{Volume} \Rightarrow Volume = \frac{Mass}{Density}$$

$$Volume = \frac{200\times10^{-3}\,kg}{\left(8\times10^{-3}\,kg\right)\times10^{6}\,m^{-3}} = 25\times10^{-6}\,m^3$$

Magnitude of magnetic moment $p_m = 2\,A\,m^2$

Intensity of magnetization,

$$I = \frac{Magnetic\,moment}{Volume} = \frac{2}{25\times10^{-6}}$$

$$M = 0.8\times10^{5}\,A\,m^{-1}$$

EXAMPLE 3.11

Using the relation $\vec{B} = \mu_\circ(\vec{H} + \vec{M})$, show that $\chi_m = \mu_r - 1$.

Solution

$\vec{B} = \mu_\circ(\vec{H} + \vec{M})$,

But from equation (3.36), in vector form, $\vec{M} = \chi_m \vec{H}$

Hence, $\vec{B} = \mu_\circ(\chi_m + 1)\vec{H} \Rightarrow \vec{B} = \mu\vec{H}$

where, $\mu = \mu_\circ(\chi_m + 1) \Rightarrow \chi_m + 1 = \frac{\mu}{\mu_\circ} = \mu_r$

$\Rightarrow \chi_m = \mu_r - 1$

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